Logic, Language, and Philosophy

Relations and orderings

Meeting 21, December 1, 2025

Martin Stokhof

Department of Philosophy, Tsinghua University, Beijing
ILLC/Department of Philosophy, University of Amsterdam

Overview

Introduction

- Reasoning with relations
- Reasoning with orderings

Relations

Orderings

Use of orderings

Homework

Introduction

Introduction

Reasoning with relations

Examples

(1) John is taller than Bill
(2) Bill is taller than Tom
(3) John is taller than Tom

(4) John is taller than everyone else except Bill
(5) Bill is taller than everyone else except John
(6) Johan and Bill are equally tall

(7) Sue is Mary's mother
(8) Mary is not Sue's mother

(9) John is Robin's brother
(10) Robin is not John's brother
(11) Robin is a woman

Example: causality

1. $\forall x \exists y Ryx$ / $\exists y \forall x Ryx$

Observation:

- Not a valid argument scheme as such
- Without further specification of R this is a counter-model:

$$1 \rightleftarrows 2$$

Now read Ryx as 'y is cause of x'; then (1) says:

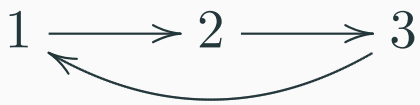
(1') Everything has a cause, therefore there is something that is the cause of everything

Note:

- Counter-model not adequate: mutual causation is strange
- Stronger: a property of causation-relation: 'if y causes x, then x does not cause y'
- But: intuitive invalidity remains, we still need a counter-model

Example: causality (2)

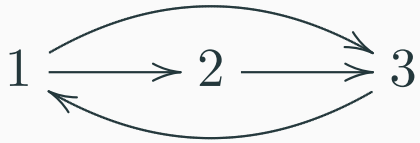
Next try:



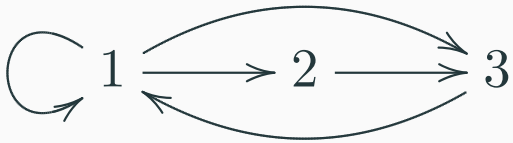
However:

- Other property of causation-relation: ‘if x causes y and y causes z, then x causes z’

But then we get:



which leads to:



But that is unacceptable because:

- ‘if y causes x, then x does not cause y’ implies ‘no x causes itself’

Causality (3)

Counter-model: $\langle \mathbb{N}, > \rangle$

Note:

- $>$ is *not* a temporal ordering
- i.e.: $x > y \neq x$ precedes y (in time)
- but: $x > y = y$ is a cause of x

Conclusion

- Relations have structural properties that (co)determine their meaning
- I.e., *relations have logical properties*

Introduction

Reasoning with orderings

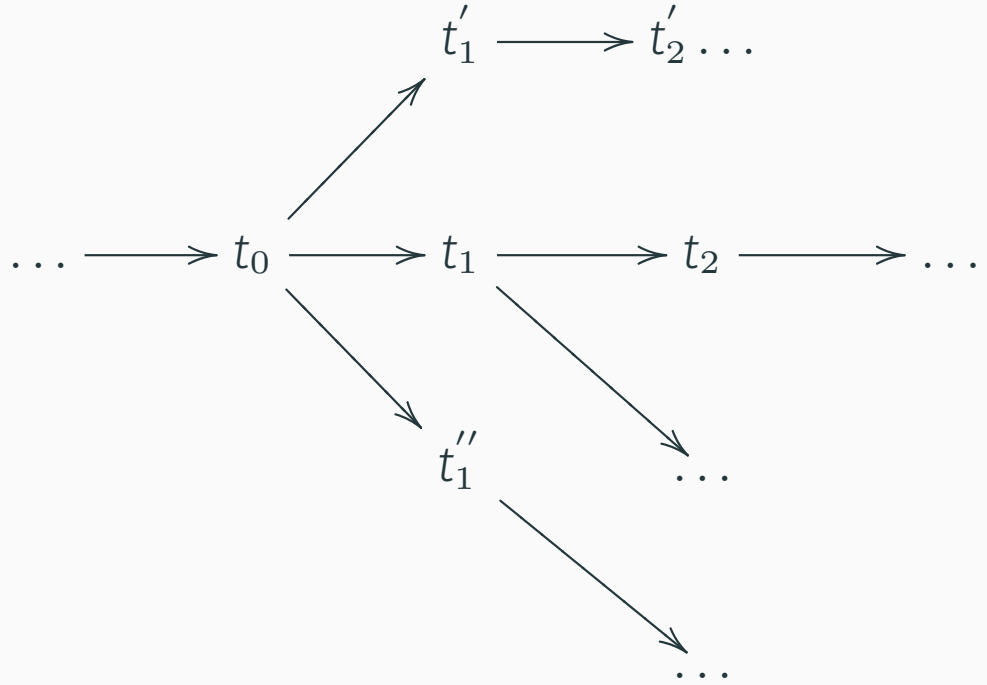
Example: time

Two models (among many):

1. Linear time:

$$\ldots \longrightarrow t_{-2} \longrightarrow t_{-1} \longrightarrow t_0 \longrightarrow t_1 \longrightarrow t_2 \longrightarrow \ldots$$

2. Branching time:



Example: time (2)

Tense operators (classical, à la Prior):

- Pp: somewhere in the past p (‘p was the case’)
- Fp: somewhere in the future p (‘p will be the case’)
- Hp: everywhere in the past p (‘p has always been the case’)
- Gp: everywhere in the future p (‘p will always be the case’)

Example: time (3)

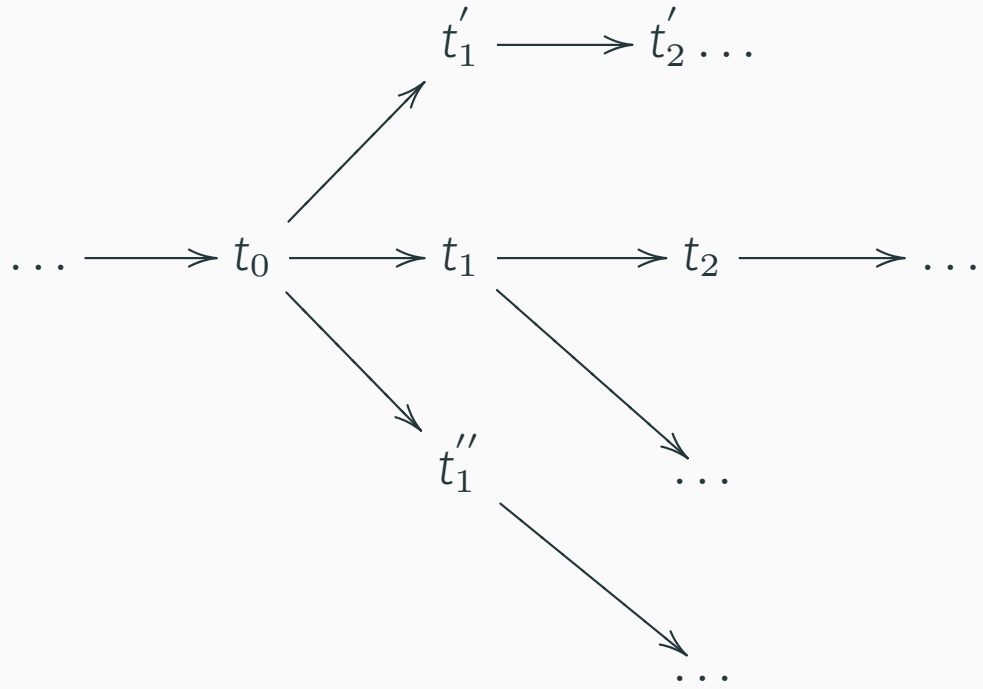
Linear time:

$$\ldots \longrightarrow t_{-2} \longrightarrow t_{-1} \longrightarrow t_0 \longrightarrow t_1 \longrightarrow t_2 \longrightarrow \ldots$$

- $p \to HFp$
- $p \to GPp$
- $Fp \to HFp$
- ...

Example: time (4)

Branching time:



- $p \to HFp$
- $p \to GPp$
- $Fp \to HFp$
- ...

Conclusion

- Certain domains have intuitive properties determine a *logical ordering of their elements*

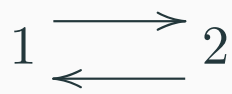
Examples:

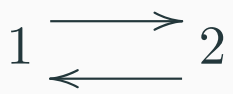
- “time has direction”: excludes circular ordering
- “future is open”: needs right-branching ordering

Relations

Symmetry

R is *symmetric* iff $\forall x \forall y(Rxy \rightarrow Ryx)$

Picture: 

or: 



Examples:

- identity
- *as tall as*
- *related by blood*

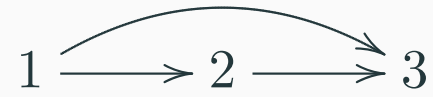
Note:

- whether a relation has a certain property may depend on the domain
  - *brother of* is not symmetrical on the domain of people
  - but it is symmetrical on the domain of all male people

Asymmetry

R is *asymmetric* iff $\forall x \forall y(Rxy \rightarrow \neg Ryx)$

Picture: 

or: 

Examples:

- *taller than*
- *parent of*

Non-symmetry

Note:

- asymmetric $\neq$ not symmetric

Diagram of a non-symmetric relation:

$$1 \longrightarrow 2 \underset{\longleftarrow}{\longrightarrow} 3$$

Examples:

- *brother of* on domain of people
- *nearest to* on points on a line:

$$x \quad y \qquad\qquad z \qquad\qquad\qquad\qquad w$$

  - y is nearest to z but z is not nearest to y because x is more near to y
  - and x is nearest to y and y is nearest to x

Anti-symmetry

R is *anti-symmetric* iff $\forall x \forall y (x \neq y \rightarrow (Rxy \rightarrow \neg Ryx))$

Picture: 

or: 

Example:

- *greater than or equal to*

Note:

- asymmetry $\models$ anti-symmetry
- equivalent definition: $\forall x \forall y ((Rxy \wedge Ryx) \rightarrow x = y)$

Reflexivity

R is *reflexive* iff $\forall x Rxx$

Picture:  or: 

Examples:

- identity
- *being of the same age/height/weight/... as*

Irreflexivity

R is *irreflexive* iff $\forall x \neg Rxx$

Examples:

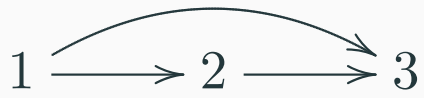
- *bigger than*
- *parent of*

Note:

- asymmetry $\models$ irreflexivity
- irreflexivity $\not\models$ asymmetry
- asymmetry $\Leftrightarrow$ anti-symmetry + irreflexivity

Transitivity

R is *transitive* iff $\forall x \forall y \forall z((Rxy \wedge Ryz) \rightarrow Rxz)$

Picture: 

Examples:

- *ancestor of*
- *greater than*

Intransitivity

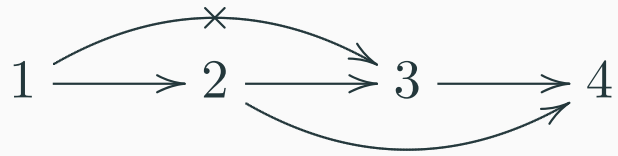
R is *intransitive* iff $\forall x \forall y \forall z((Rxy \wedge Ryz) \rightarrow \neg Rxz)$

Example:

- *mother/father/grandmother/... of*

Non-transitivity

R is *non-transitive* iff $\exists x \exists y \exists z (Rxy \wedge Ryz \wedge \neg Rxz)$



Example:

- *next to* on points on a line

Connectedness

R is *connected* iff $\forall x \forall y (x \neq y \rightarrow (Rxy \vee Ryx))$

Examples:

- $<$ on $\mathbb{N}$
- but not: *smaller than* on the domain of people

Note:

- equivalent definition: $\forall x \forall y (Rxy \vee Ryx \vee x = y)$

Combined properties

Recall the causation example. When read as 'x is a cause of y' it holds that Rxy is:

1. irreflexive: $\forall x \neg Rxx$
2. transitive: $\forall x \forall y \forall z((Ryx \wedge Rxz) \rightarrow Ryz)$
3. is serial: $\forall x \exists y Ryx$

A relation with these three properties only has models M with an infinite domain D_M.

Orderings

Preordering

R is a *preordering* (a.k.a. ‘quasi-ordering) iff R is

- reflexive
- transitive

R is a *strict preordering* iff R is

- irrreflexive
- transitive

Example:

- preference:
  - you do not prefer x to x
  - if you prefer x to y and you prefer y to z, then you prefer x to z

Partial ordering

R is a *partial ordering* iff R is

- reflexive
- transitive
- anti-symmetric

Examples:

- $\subseteq$ on the domain of sets
- $\models$ on the domain of sets of equivalent formulas

Strict partial ordering

R is a *strict partial ordering* iff *R* is

- transitive
- asymmetric

Example:

- *smaller than* on the domain of people

Note:

- asymmetry = irreflexivity + anti-symmetry
- so *R* is strict partial ordering iff *R* is:
  - irreflexive
  - transitive
  - anti-symmetric

Lattices

A *lattice* is a set with a partial ordering that meets the additional requirements that:

- every two elements have a least upper bound (“join”)
- every two elements have a least lower bound (“meet”)

Example:

- the powerset of a set, ordered by $\subset$

Linear ordering

R is a *linear ordering* iff R is

- a partial ordering
- connected

Example:

- $\leqslant$ on the natural numbers $\mathbb{N}$

Note:

- if R is a *strict* partial ordering we get *strict* linear ordering

Example:

- $<$ on $\mathbb{N}$

Equivalence relation

R is an *equivalence relation* iff R is

- reflexive
- transitive
- symmetric

Examples:

- identity
- *as big/heavy/old/... as*

Equivalence class:
Let R be an equivalence relation on X, then:

$$\{\langle x, y\rangle \mid x, y \in X \,\&\, \langle x, y\rangle \in R\}$$

is called an *equivalence class* under R

Partitions

Definition (Partition)

A *partition* of a set X is:

$$\{X_i \mid X_i \subseteq X \;\&\; X_i \neq \emptyset \;\&\; \bigcup X_i = X \;\&\; \text{for all } i, j\colon X_i \cap X_j = \emptyset\}$$

Fact

R is an equivalence relation on X iff R induces a partition of X

Use of orderings

Temporal orders

Let T be an infinite set of discrete temporal points, and Ett', read as 't is earlier than t'', a relation defined on T.

Definition (Linear time)

$\langle T, E \rangle$ is a model of *linear time* iff E is a strict linear ordering

Definition (Right-branching time)

$\langle T, E \rangle$ is a model of (right-branching) *branching* time iff

1. E is a strict partial ordering; and
2. $\forall t' \forall t'' ((Et't \wedge Et''t) \rightarrow (Et't'' \vee Et''t' \vee t' = t''))$

Note:

- t occurs free and functions as 'now'
- condition 2 states that E is connected up to now
- so conditions 1 and 2 imply that E is linear up to now

Equivalence classes and properties

Let R be an equivalence relation on D, and P a property of elements of D. Then:

Definition (Class invariant)

P is class invariant (a.k.a. 'well-defined') under R if and only if

$$\forall x \forall y (Rxy \rightarrow (Px \rightarrow Py))$$

Examples:

- $=$ on a set D defines the property of 'being d', for all $d \in D$
- *as big as* on the domain of people defines the property of *being n tall* (i.e., *having length n*)

Homework

Homework

- Read chapter 18 and make exercises 1, 3, 4(a), 6(a,c), 7(b)
- Read Russell's 'On Denoting'